AI Agents for Descriptive Social-Science Research

Instructor: Martin Alberdi
Format: Online, Zoom
Total hours: 10 scheduled hours, divided into four 2½-hour sessions (including short breaks)
Dates: 10, 12, 17, and 19 November 2026

Time: 15:30–18:00 EET

Description

This 10-hour online course introduces AI agents as practical research assistants for social scientists. Across four sessions, students use one shared descriptive question to learn agent operation, descriptive analysis, research writing, and reusable, reviewable workflows.

The course welcomes MA and PhD students with no prior programming, terminal, VS Code, or AI-agent experience. Basic data concepts are introduced in class. Students supervise agents and remain responsible for research decisions.

Learning Outcomes

By the end of the course, students will be able to:

- explain what an AI agent is and how it differs from a chatbot;
- describe how agents use context, files, tools, instructions, and skills;
- identify applications of agents to data discovery, documentation, descriptive analysis, visualization, and research writing;
- organize and supervise a descriptive research project in VS Code;
- identify official data sources and avoid unsafe or inappropriate scraping;
- document variables, missing values, and limitations;
- produce descriptive tables or visualizations relevant to the research question;
- check source references, selected values, interpretations, and agent suggestions;
- write a 200-300 word evidence brief with an AI-use disclosure.

Co-funded by the European Union

Prerequisites

Students need a laptop on which they can install VS Code, Git, Node.js, Gemini CLI with its companion extension, and Python. A browser, internet access, and eligible Gemini API access are required. Step-by-step Windows and macOS guides and a small readiness exercise are provided. Contact the instructor before class if installation or access fails; paired participation is the fallback.

Assignments

Students complete guided in-session checkpoints and one final mini-output using a shared dataset and its documentation.

Final deliverable:

- one research question answered by a checked descriptive figure;
- one 200-300 word evidence brief;
- one AI-agent supervision log recording sources, instructions, checks, and revisions.

Attendance

For certificate purposes, the recommended attendance requirement is participation in at least 3 of 4 sessions.

Course Schedule

Session 1: Getting Started with an AI Research Assistant

This session explains what agents are and how they use files, tools, and context. Students learn to navigate a project in VS Code, distinguish the terminal from the agent prompt, and specify a small task. They inspect a plan, create project instructions, and complete a supervised task. Privacy, permissions, protection of raw data, and basic verification begin here.

Session 2: From a Research Question to a Descriptive Figure

Students use a supplied dataset and official documentation to define observations, variables, percentages, and descriptive comparisons. A short demonstration introduces official data acquisition. Students supervise an analysis, inspect a figure, check a plotted value against its source, and document missing values and limitations. Saved scripts make the output repeatable; supplied data keep the exercise feasible if live access fails.

Session 3: From Results to Clear Research Writing

Students first explain their finding and its limitations in their own words. They then use agent feedback to identify unclear language, unsupported claims, and discrepancies between the figure and the text. They revise a 200–300-word evidence brief, distinguish description from causation, and record which suggestions they accepted or rejected and why.

Co-funded by the European Union

Session 4: Reusable Instructions, Verification, and Handover

Students adapt a supplied checklist for reviewing a descriptive figure and test it on another example. This introduces reusable instructions and skills without requiring advanced configuration. They review one another's project, record and address a specific issue, and assemble the checked figure, evidence brief, AI-use disclosure, and supervision log. The final discussion identifies what was verified and what remains uncertain.

Background Materials

Before session 1, read the course README, follow the Windows or macOS setup guide to clone the course GitHub repository through VS Code, and complete the readiness exercise. The first session includes a guided interface walkthrough; no advance coding study is required.

Co-funded by
the European Union